Ontology Generation using Large Language Models

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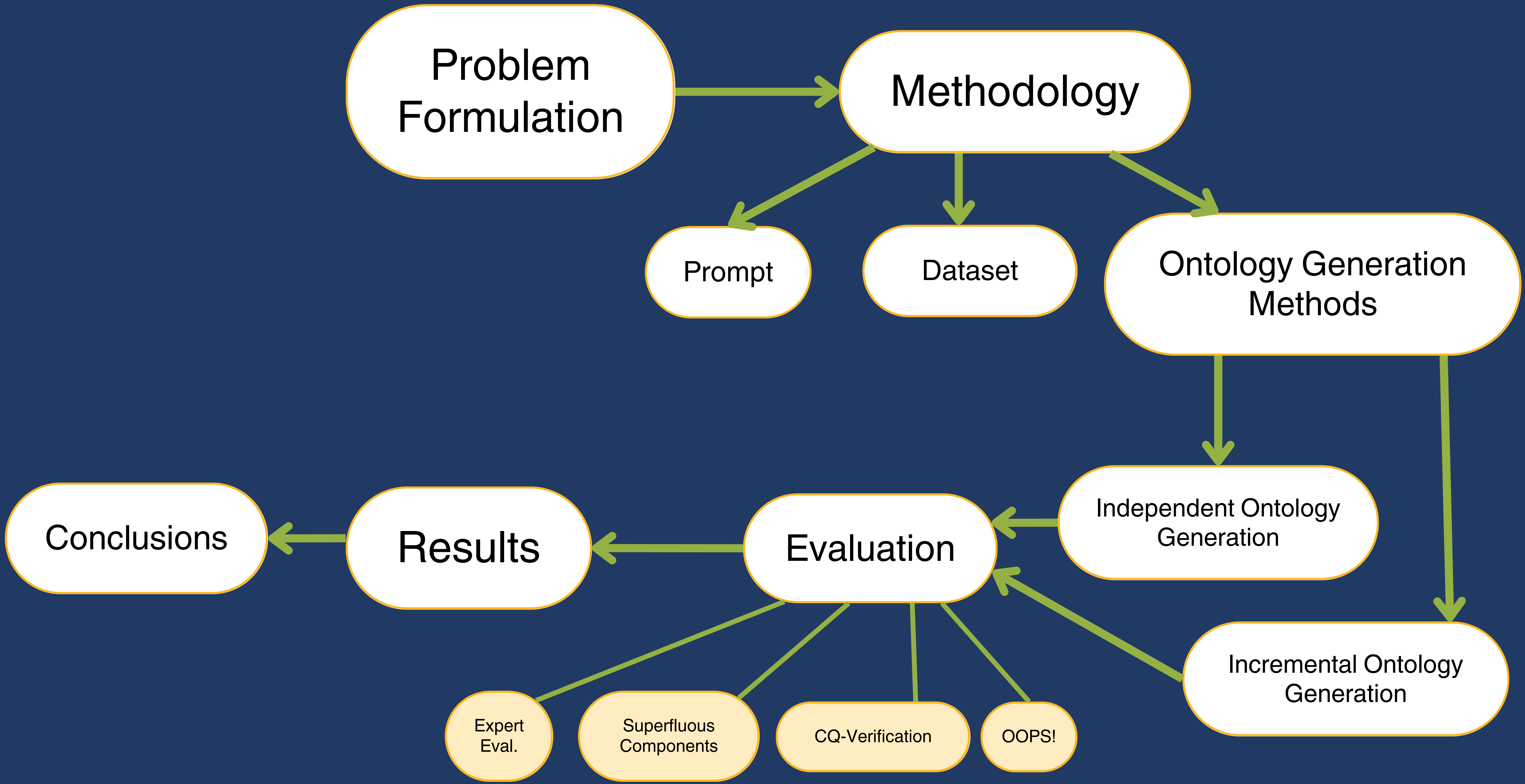
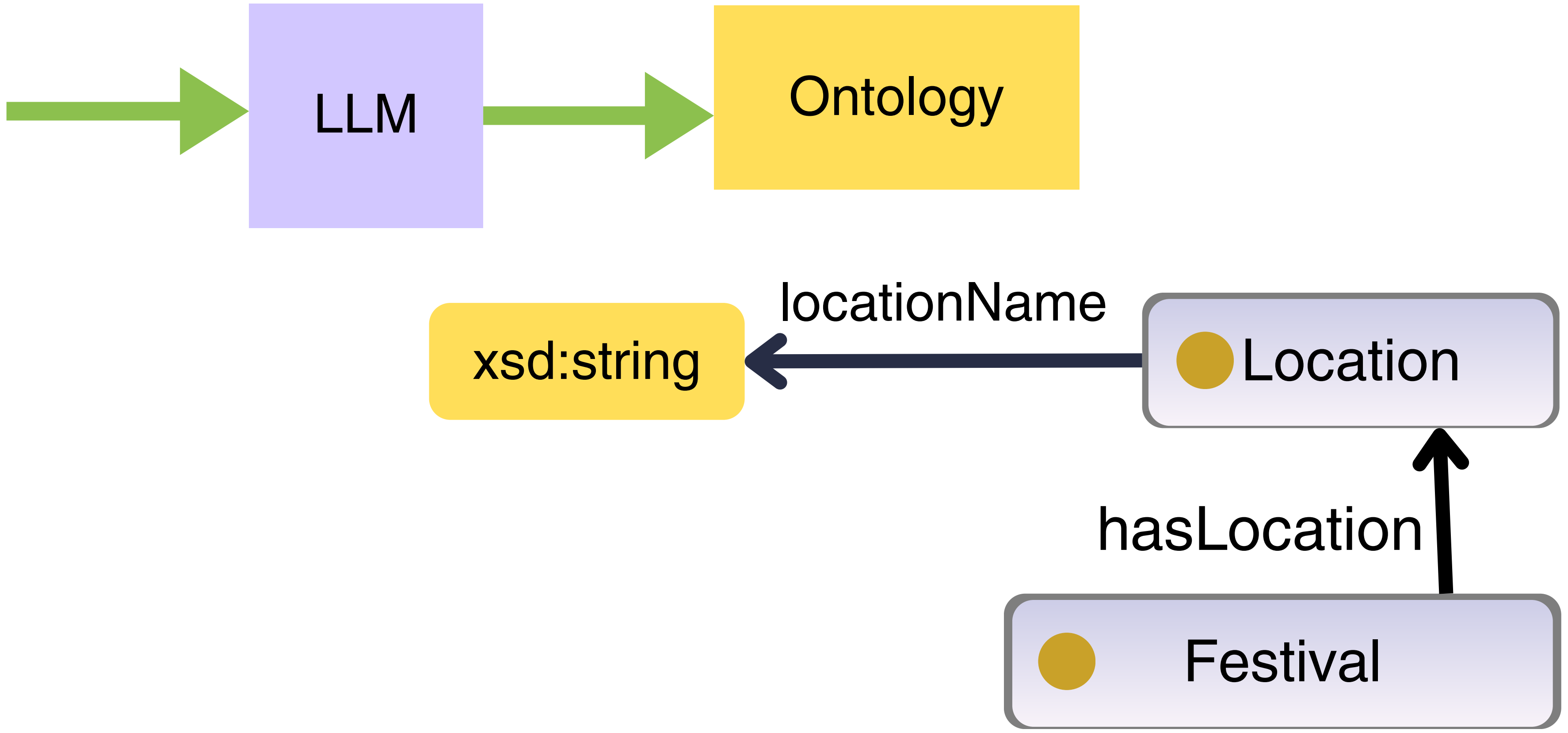
Can LLMs generate ontologies?

Map of Presentation



Problem Formulation

Requirements	Example
CQ	Where did a certain theatre festival take place?
Ontology story	The context is about music festivals happening in Italy. Each year ...



RQ 1 To what extent can LLMs be used to support the generation of ontologies that meet a predefined set of requirements?

RQ 2 What evaluation criteria are suitable for evaluating LLM-generated ontologies?

RQ 3 What are the strengths and weaknesses of ontologies generated using LLMs?

Methodology

Prompting techniques

1- Memoryless CQbyCQ

2- Ontogenia

Dataset

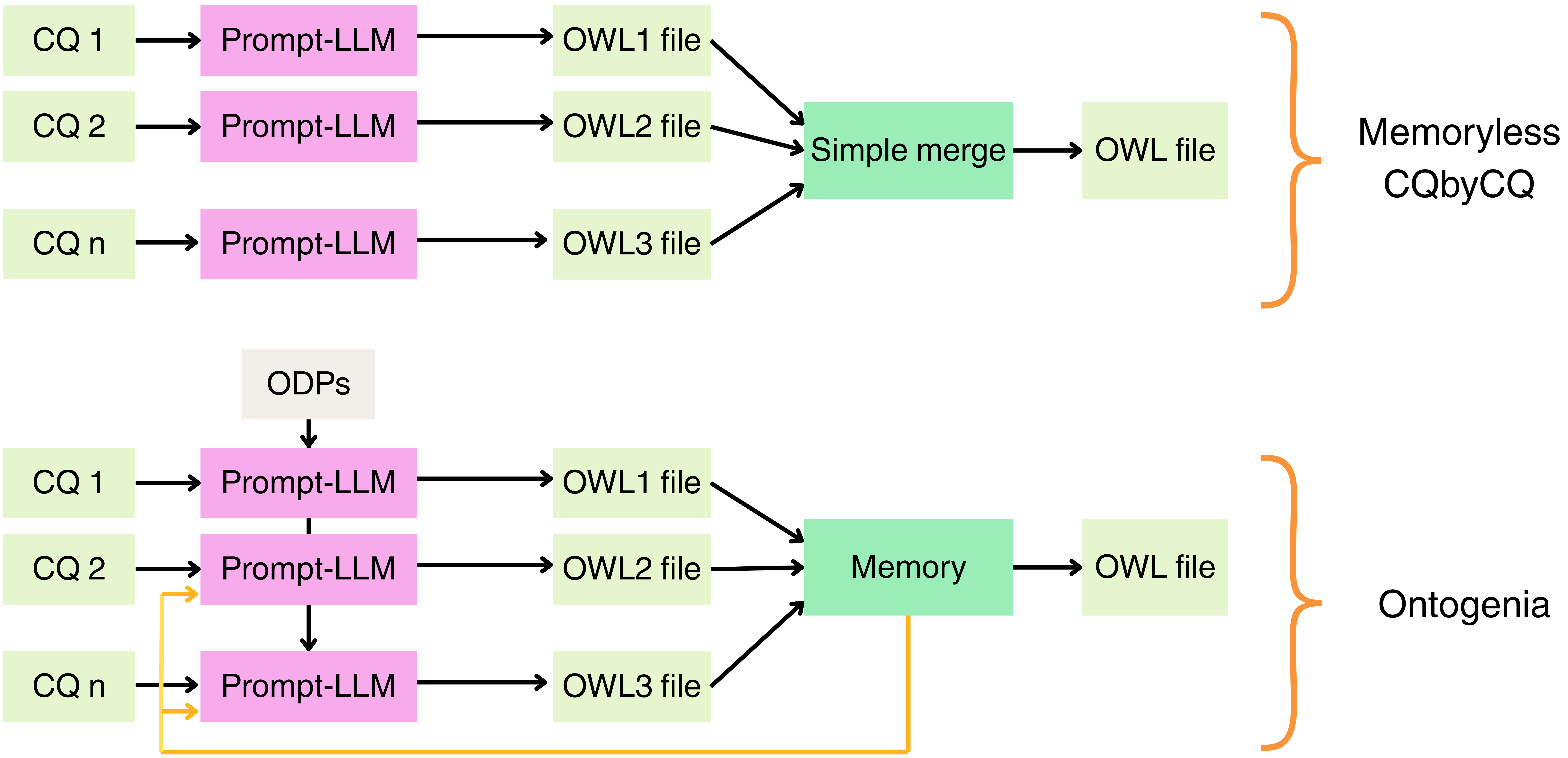
- 100 CQs
- 29 user stories
- Gold standard ontologies

Ontology Generation Methods

1- Independent Ontology Generation

2- Incremental Ontology Generation

Methodology: Prompting techniques



Methodology: Dataset

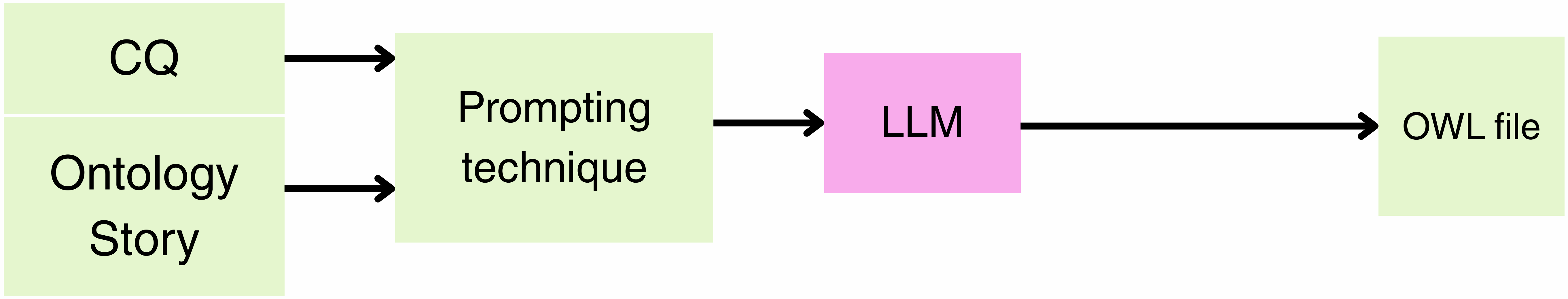
Competency Question	Story	Gold standard ontology
What are the roles of this actor in this network?	Context is about ...	CVN.ttl
What instruments does a certain person play?	Context is about ...	Music.ttl

65 complex CQs and 35 simple

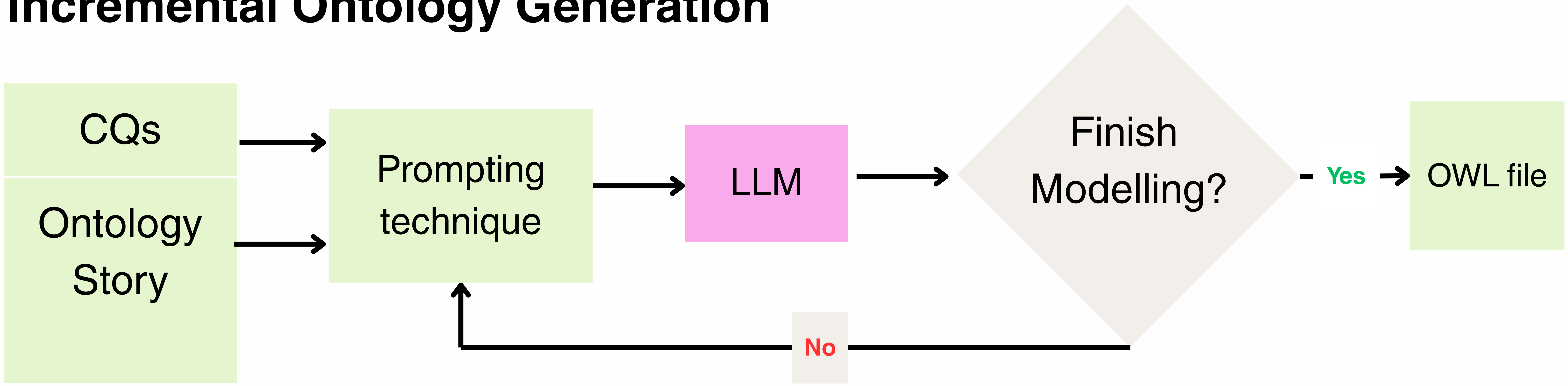
7 domains

Methodology: Ontology generation methods

1- Independent Ontology Generation



2- Incremental Ontology Generation



Evaluation

Limitation of previous work:
One-dimensional evaluation is not enough

- Based on entity label matching

Issue with synonym and with the need of a gold standard.

- Structural Statistics: E.g. axiom counting

Axiom number does not represent ontology quality.

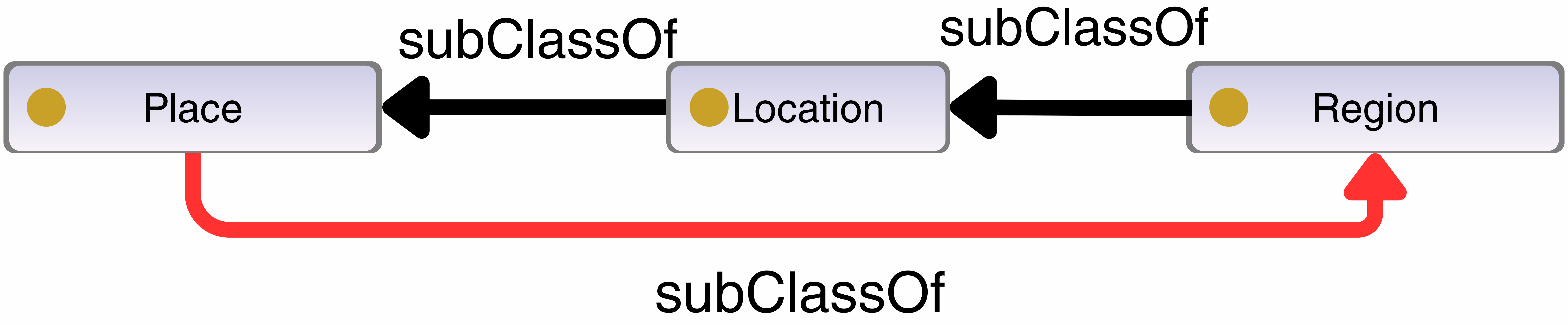
Our holistic evaluation metric

- OntOlogy Pitfall Scanner! (OOPS)
- CQ verification
- Superfluous components
- Expert evaluation

Metrics: OOPS!

OOPS!: Warns about common pitfalls in the modelling

E.g. Loops in the taxonomy.



OOPS! (OntOlogy Pitfall Scanner!): An On-line Tool for Ontology Evaluation (Poveda-Villalón et al., 2014)

Results: OOPS!

Critical Pitfalls by OOPS!

Prompting Techniques

LLMs

		MemorylessCQbyCQ			Ontogenia		
		GPT-4	Llama	o1	GPT-4	Llama	o1
P05	Wrong inverse relationships	1	25	**0**	2	7	5
P06	Cycles in a class hierarchy	**0**	2	**0**	5	**0**	11
P19	Multiple domains or ranges	23	32	1	4	15	**0**
P29	Wrong transitive relationship	**0**	**0**	1	**0**	**0**	**0**
P37	Ontology not available	2	**0**	**0**	**0**	**0**	**0**
P39	Ambiguous namespace	2	**0**	1	1	**0**	**0**

Results: OOPS!

Critical Pitfalls by OOPS!

Prompting Techniques

LLMs

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P37	Ontology not available	2	**0**	**0**	**0**	**0**	**0**
P39	Ambiguous namespace	2	**0**	1	1	**0**	**0**

Takeaways:

Llama (3.1-405 B-instruct-bf16)→ high number of pitfalls.

o1-preview → yields the least number of pitfalls

Metrics: CQ verification

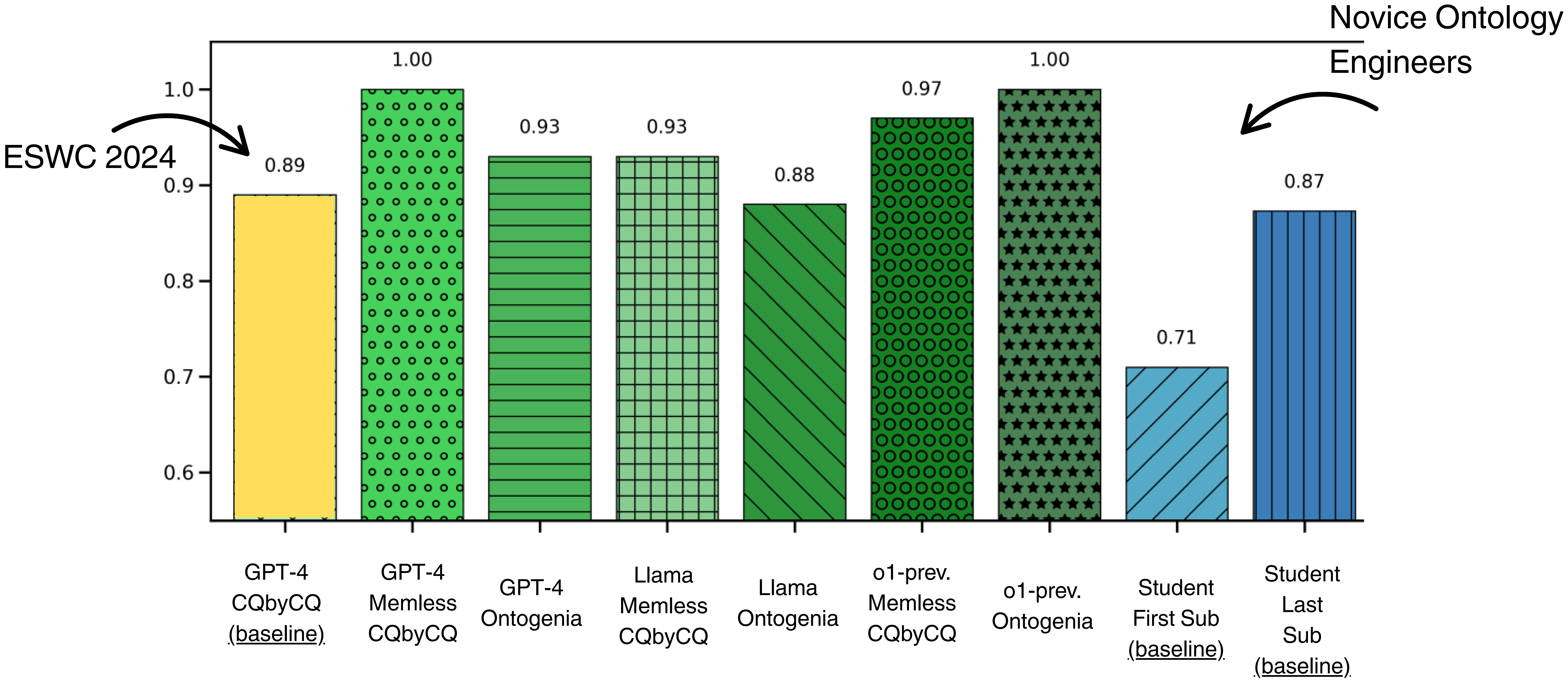
CQ verification: makes sure the ontology aligns with the intended requirements.

e.g., CQ: Who is the author of a book?

Performed mannually by two ontology engineers

Ontology testing-methodology and tool (Blomqvist et al., 2012).

Results: CQ verification



Proportion of modelled CQs

Metrics: Superfluous components

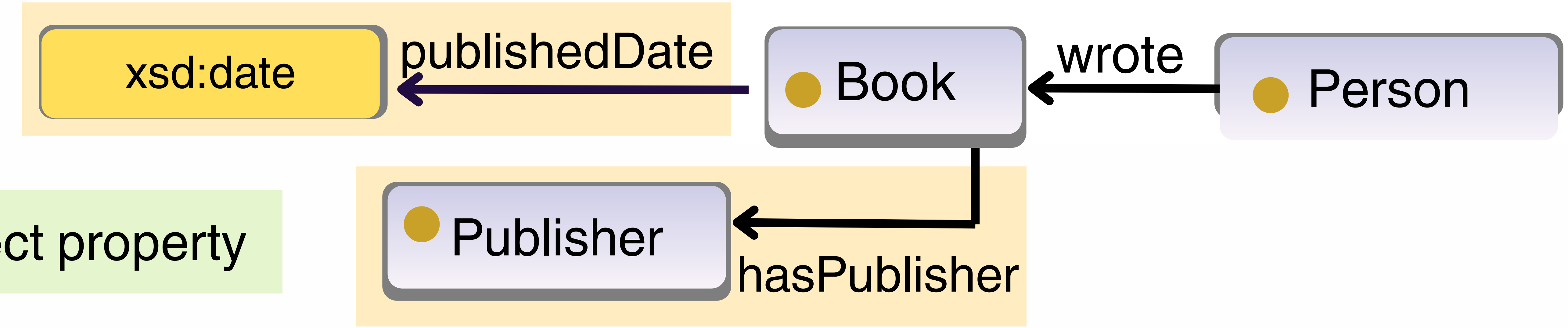
Superfluous components: Unnecessary components.

Performed manually by two ontology engineers

E.g.: CQ: Who is the author of a book?

Superfluous data property

Superfluous class and object property



Results: superfluous components

Prompting Technique	Superflu. Classes			Superflu. Obj. Properties			Superflu. Data Properties		
	Theatre	Music	Hospital	Theatre	Music	Hospital	Theatre	Music	Hospital
CQbyCQ (GPT-4) Baseline	**0**	14.3	**8.6**	**0**	**4.2**	**6.6**	**12.5**	28.5	18
Ontogenia (GPT-4)	**0**	19.2	**13.5**	**4**	38.9	29.6	45.5	55.5	22.2
Ontogenia (Llama)	47.1	32.1	37.5	40.6	45.9	38.3	16.7	**0**	100
Ontogenia (o1)	16.7	**12.5**	27.3	29.4	55.2	37	28.6	66.7	45.5
MemorylessCQbyCQ(GPT-4)	25.7	31.2	38.9	17.4	20.9	16.2	48	40.9	46.1
MemorylessCQbyCQ(Llama)	45.6	42.3	23.9	20.5	25.6	41.7	60	60.7	**13.6**
MemorylessCQbyCQ(o1)	29.3	50	28.6	**14.8**	**0**	**7.4**	**10**	**0**	28.6

Percentage of superfluous elements in each category

Takeaways:

1- Llama (3.1-405 B-instruct-bf16) generates many superfluous components, almost 40%.

2- Generally, ~30% of generated classes and properties were superfluous.

3- Considering OOPS! and CQ verification results, o1-preview is the best model so far.

Metrics: Expert Evaluation

- **Expert Evaluation**:

(i) Comments on usability, completeness and accuracy.

(ii) Assessment concerning intended modelling.

Results: Experts Evaluation

Two expert ontology engineers with the inter-annotator agreement (Cohen's kappa) of 0.61

1- No label and comments on some classes and properties.

2- Loops in the taxonomy.

3- Multiple domains and ranges for some properties.

4- Unnecessary classes.

5- Duplicate classes or properties.

6- Not all CQs are properly modelled.

7- Axioms are poorly defined for classes or no axioms at all.

8- Bad taxonomy or missing taxonomy.

Results: Experts Evaluation

1- No label and comments on some classes and properties.

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Detectable by **OOPS!**

Detectable by assessing **superfluous components**

Detectable by **CQ verification**

Conclusions

RQ1: To what extent can LLMs be used to support the generation of ontologies that meet a predefined set of requirements?

- LLMs outperform novice ontology engineers and previous methods.

RQ2: What evaluation criteria are suitable for evaluating LLM-generated ontologies?

- Several metrics <u>together</u> are required to measure the quality of generated ontologies, and mostly align with experts' evaluations.

RQ3: What are the strengths and weaknesses of ontologies generated using LLMs?

- Strengths: LLMs can generate ontology drafts & are faster than manual effort
- Weaknesses: Contain OOPS! pitfalls and superfluous elements.

THANK YOU!
QUESTIONS?

GitHub repository

Read the paper

For more questions:

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